

Exp No: 9

ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

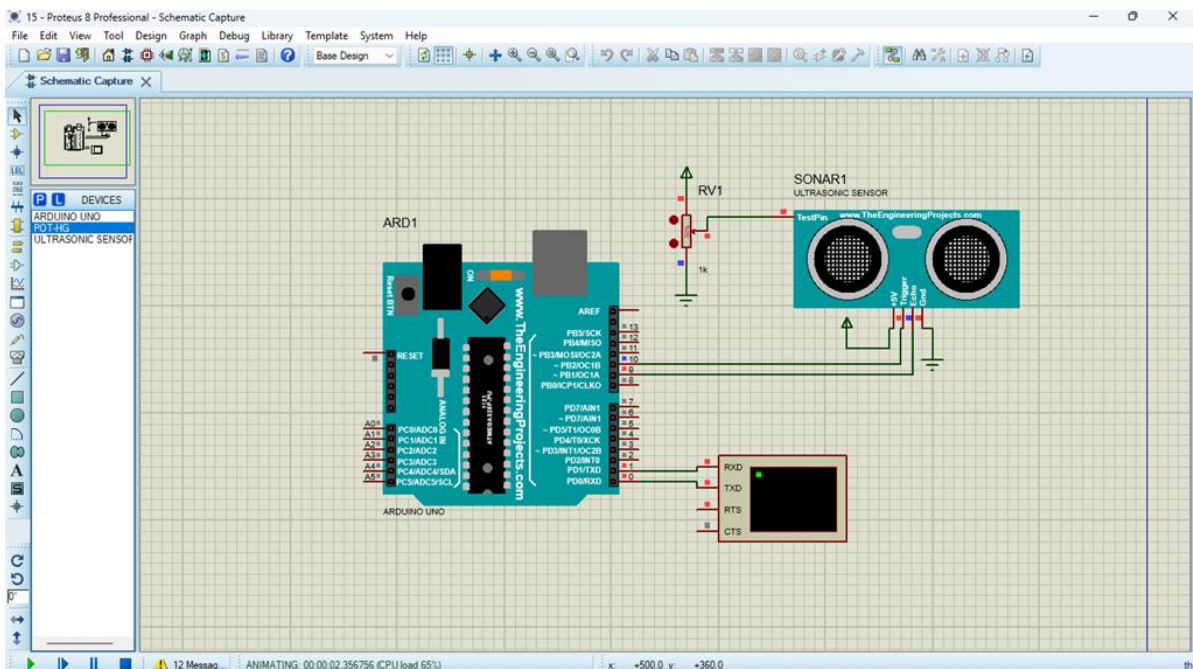
AIM:

To write an Embedded C program for interfacing Ultrasonic Sensor using Arduino Uno and Proteus

SOFTWARE REQUIRED:

- Proteus 8 software.
- Arduino IDE software

CIRCUIT DIAGRAM:



PROGRAM:

// Program Sketch

```
int trig = 10;
```

```
int echo = 9;
```

```
long duration;
```

```
int cm;
```

```
void setup() {
```

```
// put your setup code here, to run once:
```

```
pinMode(trig, OUTPUT);
```

```

pinMode(echo,INPUT);

Serial.begin(9600);

}

void loop() { // Sketch to calculate ultrasonic distance

digitalWrite(trig, LOW);

delayMicroseconds(10);

digitalWrite(trig, HIGH);

delayMicroseconds(10);

digitalWrite(trig, LOW);

delayMicroseconds(10);

duration = pulseIn(echo, HIGH);

cm = (duration/2) * 0.034;

// Print Value on Serial Monitor

Serial.print(" Distance = ");

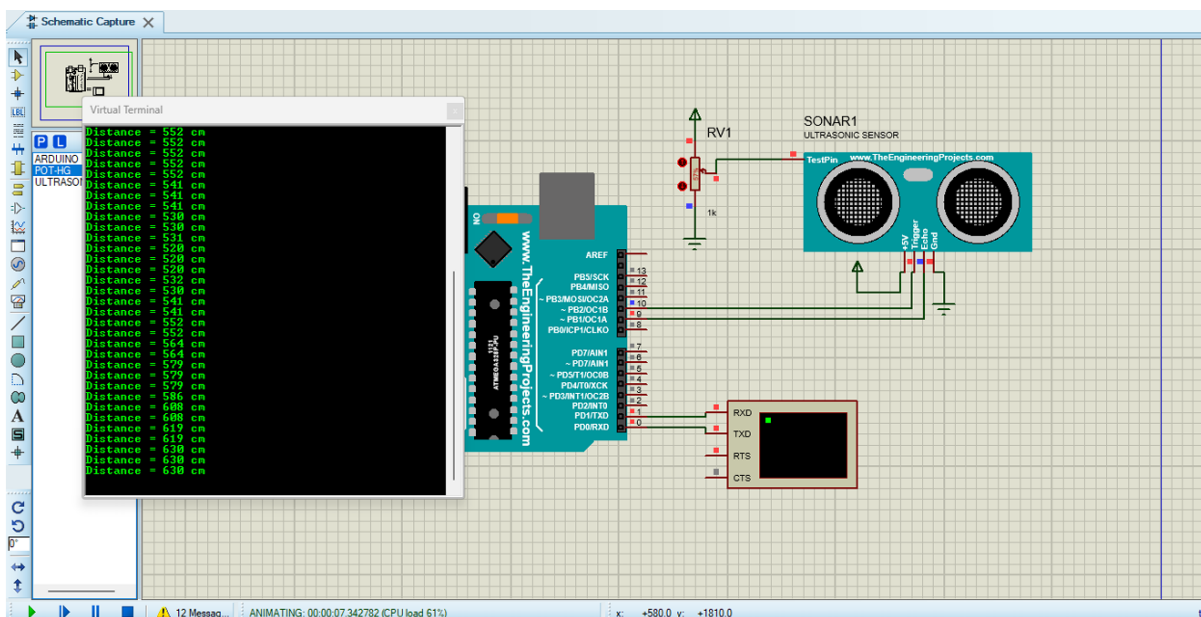
Serial.print(cm);

Serial.println(" cm");

}

```

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.